TRANSISTOR COMMON EMITTER AMPLIFIER

1. Definition

A common emitter (CE) amplifier is a transistor amplifier circuit in which the emitter terminal is common to both the input and output circuits. The input signal is applied between the base and emitter, while the output signal is taken between the collector and emitter.

It is widely used for amplifying weak electrical signals because it provides both voltage gain and current gain.

2. Circuit Components

A common emitter amplifier generally consists of the following components:

1. Transistor: Acts as the amplifying device.
2. Base resistors: Provide the required biasing voltage to the transistor.
3. Collector resistor (RC): Converts changes in collector current into changes in output voltage.
4. Emitter resistor (RE): Improves thermal stability and stabilizes the operating point.
5. Coupling capacitors: Allow AC signals to pass while blocking DC.
6. Bypass capacitor: Provides a low-impedance path for AC signals around the emitter resistor.
7. DC power supply (VCC): Supplies energy to the amplifier circuit.

3. Working Principle

The common emitter amplifier operates with the transistor in the active region.

1. A small AC input signal is applied to the base through a coupling capacitor.
2. The input signal causes small variations in the base current.
3. These variations produce larger changes in the collector current.
4. The varying collector current produces a varying voltage across the collector resistor.
5. The output voltage is obtained from the collector terminal and is amplified.
6. The output signal is 180° out of phase with the input signal.

Thus, a small input signal produces a larger output signal.

4. Voltage Gain

Voltage gain is the ratio of the output AC voltage to the input AC voltage.

Formula:

$Av = Vout / Vin$

For a simplified common emitter amplifier with an unbypassed emitter resistance:

$Av \approx -RC / RE$

The negative sign indicates a phase difference of 180° between the input and output signals. This approximation assumes the emitter resistance dominates the transistor's small-signal emitter resistance and that loading effects are negligible.

5. Characteristics

1. Provides high voltage gain.
2. Provides both current gain and power gain.
3. Produces a 180° phase shift between input and output.
4. Has moderate input resistance.
5. Has relatively high output resistance.
6. Is commonly used for small-signal amplification.

6. Advantages

1. Provides considerable voltage amplification.
2. Provides good power gain.
3. Uses a simple circuit arrangement.
4. Is suitable for amplifying weak AC signals.
5. Can be used in different electronic amplification stages.

7. Applications

1. Audio amplifiers.
2. Radio-frequency amplifier stages.
3. Signal amplification in communication systems.
4. Sensor signal-conditioning circuits.
5. Preamplifiers in electronic equipment.

8. Conclusion

The common emitter amplifier is an important transistor configuration used to amplify weak electrical signals. It provides voltage, current, and power gain, but the output signal is inverted by 180° relative to the input signal.